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28 June 1956
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MILITARY SPECIFICATION

LUMBER, SOFTWOOD, CEDAR, PORT ORFORD, AND ALASKA YELLOW, SHIP AND BOAT CONSTRUCTION

1. SCOPE

1.1 Scope. - This specification covers Port Orford and Alaska yellow cedar suitable for use in ship and boat construction.

1.2 Classification. - The cedar shall be of the following grades[1], as specified (see 6.2):

Grade A.
Grade B.
Grade C.
Grade D.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
TT-A-468 - Aluminum-Pigment; Powder and Paste, for Paint.

MILITARY
MIL-V-1174 - Varnish, Spar, Water-Resisting (Formula No. 80).
MIL-M-15176 - Mica (Extender Pigment).
MIL-W-19463 - Wood; Moisture Content Determination.

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

[1]The basis for grade determination is a piece 8 inches by 12 feet. For grades A and B, permissible defects shall be graded on the poorer face. For grades C and D, permissible defects shall be graded on the best face with the reverse face permitted to be one grade lower.

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3. REQUIREMENTS

3.1 Material. - Material shall be Port Orford cedar (Chamaecyparis lawsoniana) or Alaska yellow cedar (Chamaecyparis nootkatensis), as specified (see 6.2 and 6.6).

3.2 Dimensions. - Cedar lumber shall be uniformly manufactured to not less than the dimensions specified (see 6.2) with cross sections on the square and faces properly aligned.

3.2.1 Cedar lumber shall have parallel edges and faces subject to the standard tolerance for variations in sawing or subsequent surfacing (see 4.3.1 and 6.2).

3.3 Defects. - Except as hereinafter specified, cedar lumber shall be free of the following defects: Decay, sapwood, knots, holes, checks, cross breaks, gum spots and streaks, excessive slope of grain, abnormal lightness, insect attack, excessive moisture, pith, splits, shakes, stain, wane, warp, chipped grain, loosened grain, raised grain, torn grain, skips, bark or pitch pockets, pitch streaks, rough edges and variation in sawing.

3.3.1 The defects shall be so scattered in any one piece that no combination of them at any one point is more weakening than a single maximum permissible defect.

3.3.2 Slope of grain. - Grade A and grade B cedar lumber shall not have a slope of grain, either diagonal or spiral or combination thereof, which exceeds 1 inch in 12 inches of length (see 4.2.3). Localized deviations around permitted defects will be admitted. Grade C and grade D cedar lumber are not subject to a slope of grain restriction.

3.3.3 Insect attack. - The presence of frass or other indications of active insect attack will be cause for rejection.

3.3.4 Lightness. - Lightweight, brash cedar shall not be accepted (see 3.5).

3.4 Moisture content. - Cedar lumber shall be seasoned by air or kiln drying to moisture content of 13 ± 3 percent (see 4.4.1 and 5.2).

3.5 Specific gravity (density). - Exceptionally lightweight cedar lumber shall be rejected (see 6.4). Cedar lumber suspected of being abnormally light in weight during inspection handling shall be tested (see 4.4.2), if necessary. The specific gravity of cedar lumber so tested shall not be less than 0.40 (25.0 pounds per cubic foot density).

3.6 Grade A. -

3.6.1 Grade A cedar lumber shall be free from defects, except those specified herein, on the outside of pieces. The following defects will be admitted:

Knots - two sound tight pin knots.
Torn grain - slight.
Variation in sawing - occasional slight.
Skips - slight, on one face.
Bark pockets - two, very small; not through.
Sapwood - none - on pieces under 12 inches wide; 1 inch on pieces 12 inches and wider.

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3.7 Grade B. -

3.7.1 Grade B shall consist of all the better products of the log. Grade B cedar lumber shall be subject to the following limitations: One short split end, not longer than width of piece, in 5 percent of the pieces will be admitted. Slight torn grain or chipped grain will be admitted. In addition, the following defects, or their equivalent, will be admitted:

Knots - two sound tight pin knots.
Variation in sawing - occasional slight.
Checks - three small surface checks.
Bark pockets - three small; not through.
Skips - slight, on back of piece.
Pin wormholes - one per surface foot, extending through less than 1/4 the thickness of the piece.

In addition, grade B and better cedar lumber shall be subject to the following sapwood limitations:

1 inch on pieces under 8 inches wide.
2 inches on pieces 8 to 10 inches wide.
3 inches on pieces 10 to 12 inches wide.

3.8 Grade C. -

3.8.1 Grade C cedar lumber shall be subject to the sapwood limitations specified in 3.7.1. One short split, not exceeding the width of the piece, in 15 percent of the pieces will be admitted. In addition, four or combination of four of any of the following defects, up to the following limitations, will be admitted:

Fine shake - equal in length of width of piece.
Checks - small seasoning.
Pin wormholes - two per surface foot, extending through less than 1/4 the thickness of the piece.
Pith - 3 inches long on any surface.
Cup - 1/8 inch.
Sap stain - light, 25 percent of face.
Torn grain - medium.
Bark pockets - four, 6 inches or equivalent, slightly longer.

With one of the above, will admit one of the following, or equivalent:

Knots - two small sound, tight, or equivalent of sound tight pin. One additional small knot for each additional 2 inches in width.
Variation in sawing - occasional, slight.
Skips - occasional, slight, on edges and back.
3-inch cut-out 4 feet or more from either end in pieces 12 feet and longer in 10 percent of the shipment.

3.9 Grade D. -

3.9.1 Grade D cedar lumber will admit the following or their equivalent:

Checks - seasoning.
Cup - 1/4 inch.
Sound knots - four 1-inch.
Rough edges - provided piece is not more than 1/8 inch scant of standard finished width.
Sap stain - medium.
Sapwood - any amount.

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Variation in sawing - occasional, slight.
Skips - hit or miss.
Splits - short.
Torn grain.
Bark pockets - medium.
Wormholes - pin.
3-inch cut-out 4 feet or more from either end in pieces 12 feet and longer in 10 percent of the shipment.

3.10 End coatings. - The ends of sawn cedar lumber shall be painted with 2 coats of aluminum or mica paint. Aluminum paint shall consist of 2 pounds of aluminum powder paste, conforming to type II, class A of Specification TT-A-468, and 1 gallon of phenolic varnish, conforming to Specification MIL-V-1174. Mica paint shall consist of 2 pounds of mica conforming to type I of Specification MIL-M-15176, and 1 gallon of phenolic varnish conforming to Specification MIL-V-1174. A drying interval of 24 hours shall elapse before applying a second coat. A second alternate end coating may be any commercial end coating satisfactory to the bureau or agency concerned.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection lot. - All cedar lumber of one grade which is offered for delivery at one time and will be transported in one truck or railroad car shall be considered a lot for purposes of inspection.

4.1.1 Quantity. - The quantity of lumber in the lot shall be determined by tally, in accordance with commercial practice.

4.1.2 Sampling for dimensional inspection and moisture content. - Sample pieces of lumber shall be selected in accordance with table I for the inspection and tests specified in 4.3.1 and 4.4.1.

Table I - Sampling for dimensional inspection and moisture content
AQL = 4.0 percent (approx.) nonconforming.

Lot size pieces of lumber	Sample size pieces of lumber	Acceptance number (defective)	Rejection number (defective)
Up to 110	15	1	2
111 to 180	25	2	3
181 to 500	35	3	4
501 to 800	50	4	5
801 to 1300	75	6	7
1301 to 3200	110	8	9
3201 and over	150	11	12

4.2 Inspection. - Each piece of cedar lumber shall be visually inspected by the Government inspector to determine compliance with the requirements of this specification for species, grade[1], and absence of insect infestation.

4.2.1 Species. - The Government inspector is authorized to submit samples of lumber of doubtful species to a laboratory designated by the bureau or agency concerned for positive identification (see 6.6).

[1]Slope of grain and lightness (see 3.3) are classified as grade defects. Slope of grain and lightness shall be determined during the 100 percent visual inspection and questionable pieces shall be tested in accordance with 4.2.3 and 4.4.2, if necessary.

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4.2.2 Inspection procedure. - The Government inspector shall be present during the loading of the lumber into the highway truck or railroad car and shall, as the 100 percent grade inspection is carried out, select and examine sample pieces at random as the lumber is moved, for purposes of moisture content and dimensional determination. The number of pieces in the lot shall be tallied, and the sample required by table I shall be selected. Each sample piece shall be examined only for conformance to this specification for dimensions and moisture content specified (see 4.3.1 and 4.4.1). Any sample piece which does not meet the requirements shall be rejected, and if the number of such pieces found in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again after the contractor has removed all nonconforming pieces.

4.2.3 Slope of grain. - Slope of grain shall be determined on two adjoining sides of lumber suspected of having spiral or diagonal grain, or combination thereof.

4.3 Measurement. - Measurement of the diameter of knots or holes shall be determined by averaging the distance between lines drawn parallel to the edges and end of the piece and touching the extremities of the knot or hole being measured.

4.3.1 Dimensions. - Dimensions of both rough and surface lumber shall be subject to the standard tolerances specified for soft wood lumber.

4.4 Test procedures. -

4.4.1 Moisture content. - Moisture content shall be determined by a resistance type moisture meter in accordance with Military Specification MIL-W-19463.

4.4.2 Specific gravity. - The specific gravity shall be determined by ascertaining the ratio of the volume below the water line (see fig. 1) to that of the whole piece. The determination of specific gravity by this method shall be done as follows:

(a) Cut a piece of wood from the lumber to be tested, about 1 by 1 by 12 inches in dimension, the 12-inch dimension being along the grain.
(b) Dry the piece for 48 hours from 212° to 221°F.
(c) Dress the piece so that the cross section is rectangular and uniform throughout the length, and trim it to exactly 10 inches in length. Mark off and number the length in 1 inch demarcations.
(d) Place the piece carefully in a tall glass container of water so that it floats in an upright position. Note the water line to which its sinks. Quickly remove and mark the average wetted length (see fig. 1). The position of the water line indicates the specific gravity of the piece on the basis of oven-dry weight and oven-dry volume. Thus, if the mark is halfway between 5 and 6 inches the specific gravity if 0.55. The whole operation shall be performed quickly, because the oven-dry piece will begin to pick up moisture while it is being dressed, cut, and marked and when it is placed in the water. When determining the acceptability of suspected material, the required immersion depth may be marked on the specimens before testing.

4.5 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Narrow, thin pieces shall be securely bundled to facilitate handling.

5.2 Packing. - Unless otherwise specified in the contract or order, the loading of the lumber shall be as directed by the Government inspector. As far as practicable the lumber shall be segregated for species, grade, thickness and width. The load shall be blocked and braced to prevent shifting in transit and shall be protected against precipitation and the adverse effects of the direct rays of the sun in shipment. Cedar lumber dried to the moisture content limitations specified herein shall be shipped only in closed cars or vans and shall be stored under cover at all times.

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5.3 Marking. - In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Standard MIL-STD-129 for carload shipments. Suitable tally cards shall be conspicuously posted at access entranceways within the carrier. The tally cards shall be protected against deterioration and loss.

6. NOTES

6.1 Intended use. - This material is intended for use as solid lumber in constructing or repairing wooden craft. However, the grades specified are suitable for many other purposes. Because of its good working qualities, low shrinkage and ability to stay in place, Alaska yellow cedar is also used for patterns; because of its acid resistance it is recommended for use in acid tanks and for chemical containers. The above properties apply also to Port Orford cedar. The highly decay-resistant heartwood, the uniform texture and the ability to dry quickly after wetting makes both species particularly desirable for constructing and repairing wooden craft.

6.1.1 Grades A and B. - Grades A and B cedar lumber are suitable for use as hull planking, deck beams, shelf and clamp stringers, deck house sills, mouldings, cabinet work and interior trim of fine quality for natural, stained or enameled finishes.

6.1.2 Grade C. - Grade C cedar lumber is intended for use as interior and exterior trim, shelving, chests or closet linings, where the finest finish is not required.

6.1.3 Grade D. - Grade D cedar lumber is not suitable for use in the construction of wooden craft. Its inclusion is for the purpose of providing a grade of cedar suitable for uses approximately the same as grade C finish lumber, particularly where paint finishes are employed.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Grade and species required (see 1.2 and 3.1).
(c) Grain direction vertical or flat (see 3.3.2).
(d) Dimensions required (see 3.2).
(e) Amount of surfacing desired (see 3.2.1).
(f) That the contractor should pile material awaiting shipment on vertically aligned stickers, of uniform thickness, and protect it from precipitation and the direct rays of the sun, when directed by the Government inspector.

6.3 This specification specifies the extent of defects and sapwood (see 6.6) permitted in the poorest pieces of lumber acceptable.

6.4 Exceptionally lightweight pieces of cedar lumber have decidedly decreased strength properties. Suspected lightweight lumber should be tested, if necessary.

6.5 Because of the nonconformance to standard commercial dimensions, particularly planking lumber, of the cedar specified in this specification, it is usually necessary to manufacture it to order.

6.6 It is improtant that cedar lumber be positively identified. Port Orford cedar is readily discerned by the characteristic aromatic odor of freshly cut surfaces. Alaska yellow cedar has a rather pungent odor and the growth rings are normally very closely spaced. Identification of either species is not difficult after observing a few samples the identity of which is known.

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Custodian:
Bureau of Ships
Other interest:
None

Preparing Activity:
Navy - Bureau of Ships

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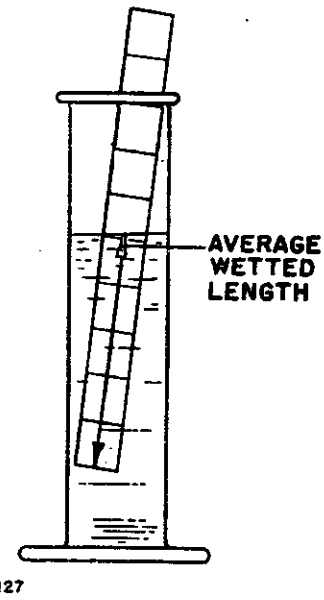


Figure 1 - Specific gravity.

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